DSA03-Natural Language Processing

List of Lab Program

13. Generate a parse tree for a given sentence using a context-free grammar using python program.

Code:

```
import nltk
from nltk import CFG
from nltk.parse import RecursiveDescentParser
grammar = CFG.fromstring("""
S -> NP VP
NP -> Det N
VP -> V N
Det -> 'the'
N -> 'boy' | 'cricket'
V -> 'plays'
""")
parser = RecursiveDescentParser(grammar)
sentence = ['the', 'boy', 'plays', 'cricket']
print("Sentence:", " ".join(sentence))
print("Parse Tree:")
for tree in parser.parse(sentence):
    print(tree)
    tree.pretty_print()
```

Output:

```
Sentence: the boy plays cricket
Parse Tree:
(S (NP (Det the) (N boy)) (VP (V plays) (N cricket)))
```

